Solutions to JEE Main - 1 | JEE - 2023

PHYSICS

SECTION-1

1.(A) Let that vector be $\vec{C}$. Then $\vec{C} = C\hat{C} = b\hat{a} \Rightarrow \vec{C} = \frac{b\vec{a}}{a} = \frac{5}{\sqrt{2}}\left(\hat{i} - \hat{j}\right)$

2.(A) For equilibrium, $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$

$(3\hat{i} + 2\hat{j} - \hat{k}) + (3\hat{i} + 4\hat{j} - 5\hat{k}) + a(\hat{i} + \hat{j} - \hat{k}) = 0$

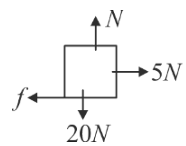
$(6+a)\hat{i} + (6+a)\hat{j} - (6+a)\hat{k} = 0$; $a = -6$

3.(A) Normal = 20 N

$(f)_{\max} = 20 \times 0.4$

$= 8N$

$f = 5N$



Force by floor on block $= \sqrt{400 + 25} = \sqrt{425}N$

4.(C) $(\vec{A} - \vec{B}) \cdot (\vec{A} \times \vec{B}) = \vec{A} \cdot (\vec{A} \times \vec{B}) - \vec{B} \cdot (\vec{A} \times \vec{B}) = 0$

As $(\vec{A} \times \vec{B})$ is perpendicular to $\vec{A}$ and $\vec{B}$

5.(B) $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1 = (\hat{i} - \hat{j} + 2\hat{k}) - (\hat{i} + \hat{j} + \hat{k}) = -2\hat{j} + \hat{k}$

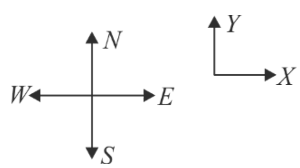
$\therefore \quad W = \vec{F} \cdot \Delta\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (-2\hat{j} + \hat{k}) = -4 + 3 = -1J$

6.(C) $\vec{S}_1 = 30\hat{j}$

$\vec{S}_2 = 20\hat{i}$

$\vec{S}_3 = -30\hat{i} - 30\hat{j}$

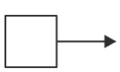
$\vec{S} = \vec{S}_1 + \vec{S}_2 + \vec{S}_3 = -10\hat{i}$



7.(A) $|\vec{A} \times \vec{B}| = \sqrt{3}\,\vec{A}.\vec{B}$

$\Rightarrow AB\sin\theta = \sqrt{3}AB\cos\theta$ or $\tan\theta = \sqrt{3} \Rightarrow \theta = 60°$

$\left|\vec{A} + \vec{B}\right| = \sqrt{A^2 + B^2 + 2AB\cos 60°} = \sqrt{A^2 + B^2 + AB}$

8.(A)  $10\cos 60 = 10 \times \frac{1}{2} = 5N$

9.(A) $T_2 \cos 60° = Mg$,

i.e., $T_2 = 2Mg$ … (i)

and for horizontal equilibrium of P

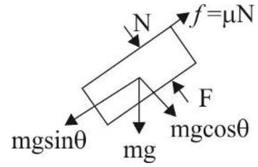
$T_1 = T_2 \sin 60° = T_2(\sqrt{3}/2)$ … (ii)

Substituting the value of T_2 from equation (i) in (ii),

$T_1 = (2Mg) \times (\sqrt{3}/2) = (\sqrt{3})Mg$

10.(D) $N = F - mg\cos\theta$ ....(i)

$f - mg\sin\theta = 0$ ....(ii)



For limiting case, force applied is minimum.

$$\mu(F - mg\cos\theta) = mg\sin\theta$$

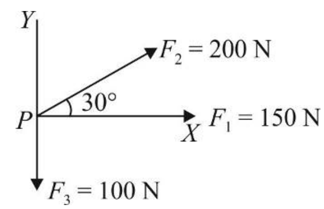
$$F = \frac{mg\sin\theta}{\mu} + mg\cos\theta$$

$$F = mg\left(\frac{\sin\theta}{\mu} + \cos\theta\right)$$

$$F = 100\left(\frac{\sin 37^\circ}{1/2} + \cos 37^\circ\right); \qquad F = 100\left(2\times\frac{3}{5} + \frac{4}{5}\right) = 200N$$

11.(C) Let applied force is $F_4 = F_x\hat{i} + F_y\hat{j}$

To prevent the particle P from moving, net force on the particle will be zero.



$\therefore \quad \sum F_x = 0$

or $\quad \sum F_x = F_x + 150 + 200\cos 30^\circ = 0$

$\therefore \quad F_x = -150 - 100\sqrt{3} = -(150 + 170) = -320N$

and $\quad F_y = 0$

It means $\vec{F}_4$ is directed opposite to $\vec{F}_1$.

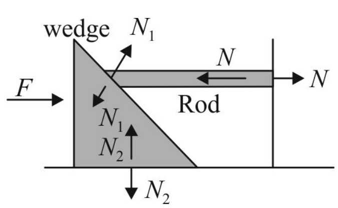
12.(C) $\vec{a}\times\vec{b}$ is perpendicular to the plane of $\vec{a}$ and $\vec{b}$

unit vector along $\vec{a}\times\vec{b}$,

$$\therefore \quad \hat{n} = \frac{\vec{a}\times\vec{b}}{|\vec{a}||\vec{b}|} = \frac{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & 1 \\ 1 & 2 & 2 \end{vmatrix}}{(\sqrt{4+4+1}\sqrt{1+4+4})}$$

$$= \frac{\hat{i}(-4-2) - \hat{j}(4-1) + \hat{k}(4+2)}{9} \qquad = -\frac{6}{9}\hat{i} - \frac{3}{9}\hat{j} + \frac{6}{9}\hat{k} = -\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$$

13.(D)



Taking Rod + wedge as a system

Balancing force in horizontal direction.

$N = F$

14.(D) $\vec{X} + \overrightarrow{BD} = \vec{Z}$

$\vec{Z} + \overrightarrow{DC} = \vec{Y}$

$\Rightarrow \quad$ As $\overrightarrow{BD} = \overrightarrow{DC}$

$\therefore \quad \vec{Z} + (\vec{Z} - \vec{X}) = \vec{Y} \quad \Rightarrow \quad \vec{X} + \vec{Y} = 2\vec{Z}$

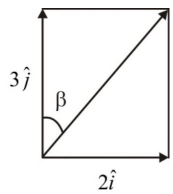
15.(B) Area of parallelogram: $\left|\vec{A}\times\vec{B}\right| = AB/2$ (given)

$\Rightarrow\ AB\sin\theta = AB/2 \quad \Rightarrow\ \sin\theta = 1/2 \Rightarrow \theta = 30°$

16.(B) $\vec{R}=\vec{A}+\vec{B}=12\hat{i}+5\hat{j}$

$R=\left|\vec{R}\right|=\sqrt{(12)^2+(5)^2}=13\,; \qquad \hat{R}=\dfrac{\vec{R}}{|\vec{R}|}$

17.(C)



$\tan\beta=\dfrac{2}{3}$ or $\beta=\tan^{-1}\left(\dfrac{2}{3}\right)$

18.(A) Component of A along B

$=A\cos\theta \qquad =\dfrac{\vec{A}.\vec{B}}{|\vec{B}|}=\dfrac{2+3}{\sqrt{1+1}}=\dfrac{5}{\sqrt{2}}$

19.(C) Given horizontal force F = 25 N and coefficient of friction between block and wall (μ) = 0.4

We know that at equilibrium horizontal force provides the normal reaction to the block against the wall. Therefore, normal reaction to the block (R) = F = 25 N.
We also know that weight of the block (W) = Frictional force
$=\mu R=0.4\times 25=10N.$

20.(A) Only two forces are acting, mg and net contact force (resultant of friction and normal reaction) from the inclined plane. Since the body is at rest. Therefore these two forces should be equal and opposite.

SECTION-2

21.(2) Given $R=A=B,$ it will give $\theta=120°$

22.(1) If $\vec{a}$ and $\vec{b}$ are perpendicular, $\vec{a}\cdot\vec{b}=0$
or $\quad 2x-3+1=0 \quad \therefore \quad x=1$

23.(7) Power, $P=F\cdot v$

$=(\hat{i}+\hat{j}+\hat{k})\cdot(3\hat{i}-\hat{j}+5\hat{k})=3-1+5=7\text{W}$

24.(5) $v=100$ m/s

Let $v_x=50\sqrt{3}$ m/s $=v\cos\theta$

$\Rightarrow \quad 100\cos\theta=50\sqrt{3} \qquad$ or $\qquad \cos\theta=\dfrac{\sqrt{3}}{2}$

$\Rightarrow \quad \theta=30°$

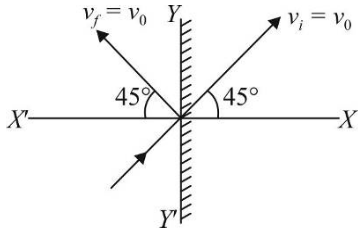
$v_y=v\sin\theta=100\sin 30°=100\times\dfrac{1}{2}$

$=50$ m/s

$=(10\times 5)$ m/s $\quad \therefore \quad x=5$

25.(2) Initial velocity is $v = v_0 \cos 45° \hat{i} + v_0 \sin 45° \hat{j}$

$$= 5\sqrt{2} \times \frac{1}{\sqrt{2}} \hat{i} + 5\sqrt{2} \times \frac{1}{\sqrt{2}} \hat{j} = 5\hat{i} + 5\hat{j}$$



The final velocity

$$v_f = -v_0 \cos 45° \hat{i} + v_0 \sin 45° \hat{j}$$

$$= -5\sqrt{2} \times \frac{1}{\sqrt{2}} \hat{i} + 5\sqrt{2} \times \frac{1}{\sqrt{2}} \hat{j} = -5\hat{i} + 5\hat{j}$$

$\therefore$ The change in velocity is

$$\Delta v = v_f - v_i$$

$$= (-5\hat{i} + 5\hat{j}) - (5\hat{i} + 5\hat{j}) = -10\hat{i} \qquad \therefore \qquad |\Delta v| = 10 \text{ ms}^{-1}$$

Chemistry

SECTION-1

1.(C) Volume of 350 drops = 20 mL

$\therefore$ Volume of 1 drops $=\dfrac{20}{350}$ mL

Mass of 1 drop = volume × density $=\dfrac{20}{350}\times 1.2\text{ g}=\dfrac{2.4}{35}\text{g}$

Number of moles in 1 drop $=\dfrac{2.4}{35}\times\dfrac{1}{70}=\dfrac{1.2}{(35)^2}$

Number of molecules $=\dfrac{1.2}{(35)^2}N_A$

2.(C) $\underset{1\text{ mol}}{4A}+\underset{0.6\text{ mol}}{2B}+\underset{0.72\text{ mol}}{3C}\longrightarrow A_4B_2C_3$

1 mol A requires 0.5 mol B and 0.75 mol C so C is the limiting reagent. 3 mol C gives 1 mol product

0.72 mol C will give $\dfrac{1}{3}\times 0.72$ mol product = 0.24 mol product

3.(B) Total meq of HCl added $=0.5\times 10=5$

meq of excess HCl = meq of NaOH = 0.2 × 10 = 2

meq of HCl used by $Ba(OH)_2$ = 3 = meq of $Ba(OH)_2$

$$\text{meq}=\frac{g}{E}\times 1000$$

$$3=\frac{g}{171/2}\times 1000 \;\Rightarrow\; g=\frac{3\times 171}{2}\times\frac{1}{1000}=\frac{256.5}{1000}$$

% of $Ba(OH)_2$ in the sample $=\dfrac{256.5}{1000\times 20}\times 100=1.28\%$

4.(D) **I.** Molality and mole fraction are independent of temperature as number of moles will not be affected by change of temperature.

II. Correct

$$m=\frac{M}{1000d-MM_0}\times 1000$$

If $d=1\text{gm}^{-1}$

$$m=\frac{M}{1000-MM_0}\times 1000$$

For very dilute solution, $1000-MM_0\simeq 1000$

$$m=\frac{M}{1000}\times 1000$$

III. Specific gravity is dimensionless quantity. It is equal to the ratio of density of substance and density of water.

5.(B) $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$

Mass of $H_2SO_4 = 7 \times 2 = 14g$

Pure $H_2SO_4 = \frac{7}{100} \times 14 = 0.98g$ H_2SO_4 is used

$n_{H_2SO_4} = \frac{0.98}{98} = 0.01 =$ moles of H_2

Volume of H_2 at STP $= 0.01 \times 22.4 = 0.224L$

6.(C) 40 g NaOH in 100 ml solution

Mass of solution $= 100 \times 1.12 = 112\,g$

Mass of solvent $= 112 - 40 = 72g$

Molality $= \frac{1}{72} \times 1000 = 13.8$ molal

7.(B) meq of $H^+ = (20 \times 1) + (30 \times 2 \times 2) = 20 + 120 = 140$

meq of $OH^- = 30$ $\therefore$ meq of H^+ in final solution = 110

Normality of $H^+ = \frac{110}{60} = 1.83N$

8.(B) $MgCO_3 \xrightarrow{\Delta} \underset{8g}{MgO} + CO_2$

Moles of MgO $= \frac{8}{40}$ mol $= \frac{1}{5}$ mol

$\frac{1}{5}$ mol $MgCO_3$ will give $\frac{1}{5}$ mol MgO $\therefore$ Mass of $MgCO_3 = \frac{1}{5} \times 84g$

% Purity $= \frac{84}{5} \times \frac{1}{20} \times 100 = 84\%$

9.(A)

$$M_0(x) + (2 \times M_0(y)) = 100\text{ g}$$
$$(3 \times M_0(x)) + (2 \times M_0(y)) = 180g$$
$$2M_0(x) = 80\text{ g}$$
$$M_0(x) = 40\text{ g}$$
$$M_0(y) = 30\text{ g}$$

10.(A) Let mmol of $Na_2CO_3 = x$ and $NaHCO_3 = y$ in 10 mL of solution

At phenolphthalein end point :

Meq of acid $= \frac{1}{2}$ meq of Na_2CO_3

$0.1 \times 2 \times 2.5 = \frac{1}{2} \times 2x \Rightarrow 0.5 = x$

At methyl orange end point :

meq of acid $= \frac{1}{2}$meq of $Na_2CO_3 +$ meq of $NaHCO_3$

$0.1 \times 2 \times 5 = \left(\frac{1}{2} \times 2x\right) + y$; $1 = x + y \Rightarrow y = 0.5$

Strength of $Na_2CO_3 = \frac{0.5}{10} \times 106 = 5.3$

11.(D) $NaI + AgNO_3 \longrightarrow AgI + NaNO_3$

$2AgI + Fe \longrightarrow FeI_2 + 2Ag$

$2FeI_2 + 3Cl_2 \longrightarrow 2FeCl_3 + 2I_2$

2 mol of $I_2 \equiv$ 2 mol $FeI_2 \equiv$ 4 mol $AgI \equiv$ 4 mol $AgNO_3$

1 mol of $I_2 \equiv$ 2 mol $AgNO_3$

So $\frac{1000}{254}$ mol of $I_2 \equiv 2 \times \frac{1000}{254}$ mol of $AgNO_3$

$$= 2 \times \frac{170}{254} \times 1000\,\text{g } AgNO_3 = 1338.58\,\text{g} \approx 1.34\,\text{kg}$$

12.(B) $KClO_3 \longrightarrow KCl + \frac{3}{2}O_2$

0.384 g of O_2 is lost

0.012 mol of O_2 is lost

So $\frac{2}{3} \times 0.012$ mol $KClO_3$ must be consumed

Mass of $KClO_3$ consumed $= \frac{2}{3} \times 0.012 \times 122.5 = 0.98\,\text{g}$

% of $KClO_3$ consumed $= \frac{0.98}{4.9} \times 100 = 20\%$

13.(B)

	HNO_3	+ KOH	$\longrightarrow$ KNO_3	+ H_2O
	80 ml	120 ml		
	0.2 M	0.150 M		
mmol	16	18		
	–	2	16	16

$$[K^+] = \frac{\text{Total mmol of } K^+}{\text{Total volume}} = \frac{2+16}{200} = \frac{18}{200} = 0.09$$

$$[NO_3^-] = \frac{16}{200} = 0.08$$

14.(D) meq of $H_2SO_4 = 0.1 \times 100 = 10$

meq of $KOH = 0.1 \times 40 = 4$

meq of H_2SO_4 left after reaction $= 10 - 4 = 6$

For excess of H_2SO_4, now $Ca(OH)_2$ is used.

meq of remaining $H_2SO_4 =$ meq of $Ca(OH)_2$

$$6 = 0.5 \times 2 \times V \quad \Rightarrow \quad V = 6\,\text{ml}$$

15.(D) $SnO_2 \;+\; 2H_2 \;\rightarrow\; Sn \;+\; 2H_2O$

2gm $\quad\Downarrow$

$\frac{2}{151}$ mol $\quad \frac{4}{151}$ mol

$\Downarrow$

$\frac{4}{151} \times 22.4\,\text{lit.}$

$= 0.593\,\text{lit.}$

16.(A) $C_7H_6O_3 + C_4H_6O_3 \longrightarrow C_9H_8O_4 + C_2H_4O_2$

1gm

$\Downarrow$

$$\frac{1}{138.12}\text{mol} \longrightarrow \frac{1}{138.12}\text{mol}$$

$$= \frac{1}{138.12} \times 180.15 = 1.304\,\text{gm}$$

$$\%\text{ yield} = \frac{0.85}{1.304} \times 100 = 65.18\%$$

17.(A) mmol = M × VmL

mmol of $[Na^+]$ from $NaNO_3$ = 0.1 × 50 = 5 mmol

mmol of $[Na^+]$ from Na_2CO_3 = 0.1 × 25 × 2 = 5 mmol

Total mmol of $[Na^+]$ = 10 mmol

Total volume = (50 + 25)mL = 75 mL

$$\therefore \quad \text{molarity of } Na^+ = \frac{\text{mmol of } Na^+}{V_{mL}} = \frac{10}{75}\text{mol } L^{-1} = 0.133 \text{ mol} L^{-1}$$

18.(D) $\text{Molarity} = \dfrac{10xd}{M_0}$ (x is the percentage d is the density)

$$= \frac{10 \times 38 \times 1.29}{98} = 5.00 \text{ M}$$

19.(C) $\text{Eq. wt. of acid} = \dfrac{0.1176}{15 \times 0.24} \times 1000 = 32.66 \quad \Rightarrow \quad \text{Basicity} = \dfrac{M}{E} = \dfrac{98}{32.66} = 3$

20.(D) $\text{Mol of } O_2 = \dfrac{44.8}{22.4} = 2.0 \qquad \text{Mol of } CH_3OH = \dfrac{64}{32} = 2.0$

$\because$ 2 mol CH_3OH require 3 mol O_2, hence O_2 is a limiting reagent.

Moles of CO_2 formed $= \dfrac{2}{3} \times 2 = 1.33$

SECTION-2

21.(8) Let the sample contain 'a' g chalk and $(5-a)$g clay.

On heating : Clay loses water and $CaCO_3$ loses CO_2

Loss of water by clay $= \dfrac{15 \times (5-a)}{100}$ g

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2 \uparrow$$

Loss of CO_2 by $CaCO_3$ = Number of moles × M_0 of CO_2

Number of moles of CO_2 = Number of moles of $CaCO_3 = \dfrac{g}{M_0} = \dfrac{a}{100}$

So, mass of $CO_2 = \dfrac{a}{100} \times 44$ g

Total loss in mass = 1.91g

$$\frac{15 \times (5-a)}{100} + \frac{a \times 44}{100} = 1.91\text{g}; \qquad \frac{75 + 29a}{100} = 1.91 \quad \Rightarrow \quad a = 4$$

$$\%\text{ of chalk} = \frac{4}{5} \times 100 = 80\%; \qquad x = 8$$

22.(6) $Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(s) + 3CO(g)$

1 mol of Fe_2O_3 requires 3 mol of C

So 6 mol of Fe_2O_3 will require 18 mol of C

But only 9 mol of C is present.

So the limiting reagent is carbon

3 mol of C gives 2 mole of Fe

So 9 mol of C will give 6 mol of Fe

$\therefore$ theoretical yield of Fe = 6 mol

23.(0) $AgNO_3 + KI \longrightarrow AgI\downarrow + KNO_3$

4	12	-	-
-	8	4	4

At the end of reaction, AgI gets pptd,

So no Ag^+ left in solution

$\therefore \quad [Ag^+] = 0$

24.(4) Mass of nitrogen in 194 u caffeine $= \frac{28.9}{100} \times 194 = 56u$

$\therefore \quad$ number of N atoms $= \frac{56}{14} = 4$ in 1 molecule

25.(4) Let molarity of HNO_3 be M

meq of acid $= 600\,M$

meq of $Ca(OH)_2 = 200 \times 2 \times M = 400\,M$

As solution is acidic,

meq of acid $-$ meq of base $=$ meq of excess acid $= 200\,M$

meq of excess acid $= 200\,M$

Concentration of acid $= \frac{200\,M}{V_{total}} = \frac{200\,M}{800}$

$$\frac{200\,M}{800} = 10^{-2} \quad \Rightarrow \quad M = 4 \times 10^{-2}$$

Mathematics

SECTION - 1

1.(C) We have, $|x-3|^2+|x-3|-2=0$

$\Rightarrow \quad |x-3|=1 \quad \Rightarrow \quad x-3=\pm 1 \Rightarrow x=2, 4$

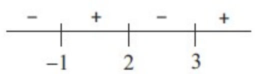
2.(D) One root $=\dfrac{1}{1+\sqrt{3}}=\dfrac{1-\sqrt{3}}{-2}=-\dfrac{1}{2}+\dfrac{\sqrt{3}}{2}$ $\therefore$ Other root $=\dfrac{-1}{2}-\dfrac{\sqrt{3}}{2}$

Sum = – 1, product $=\dfrac{1}{4}-\dfrac{3}{4}=-\dfrac{1}{2}$

$\therefore$ Required equation is $x^2+x-\dfrac{1}{2}=0$ or $2x^2+2x-1=0$

3.(D) $x^2+4x+12=9 \Rightarrow x^2+4x+3=0 \Rightarrow x=-1, -3$

4.(A) For y to be real $\dfrac{(x+1)(x-3)}{x-2}\geq 0$

$\Rightarrow$ 

Now $f(x)\geq 0 \forall x\in[-1, 2)\cup[3, \infty)$

5.(D) $a>0;\ f(0)=c>0; \dfrac{-b}{a}>0 \quad \therefore\ -b>0 \Rightarrow b<0$

Hence $abc<0$ and $D=b^2-4ac>0$

6.(C) $\alpha(\alpha\beta)^5=32 \Rightarrow \alpha=1\ [\because \alpha\beta=2]$

The value of α will satisfy the given equation $1+a+2=0 \Rightarrow a=-3$

7.(A) $f(0)<0 \Rightarrow k^2-3k+4<0$

which is a quadratic in k and $D<0\ \therefore\ k^2-3k+4>0\ \ \forall k\in R.$

$f(0)<0$ not possible.

8.(C) $m=\dfrac{-b}{2a}=1,\ n=\dfrac{-D}{4a}=4$

Now check alternative for parabola having vertex $\left(2m, \dfrac{n}{2}\right)$ i.e., $(2, 2)$

9.(B) We have $(x+1)^2>5x-1 \Rightarrow x^2-3x+2>0$

$\Rightarrow\ x<1$ or $x>2$ ....(1)

$(x+1)^2<7x-3 \Rightarrow x^2-5x+4<0 \Rightarrow 1<x<4$ ....(2)

(1) and (2) $x\in(2, 4)$ $\therefore$ Only one integral value of x which is 3.

10.(C) $x^2+px+q-p=0;$ $1-p$ is root

$(1-p)^2+p(1-o)+1-p=0$

$(1-p)(1-p+p+1)=0;$ $p=1$

$x^2+x=0 \begin{matrix} 0 \\ -1 \end{matrix}$

11.(C) $mx^2+3x+4<5(x^2+2x+2) \quad \Rightarrow \quad (m-5)x^2-7x-6<0\ \forall x \in R$

$\Rightarrow \quad m-5<0$ and $(-7)^2+24(m-5)<0 \quad \Rightarrow \quad m<5$ and $24m-71<0 \Rightarrow \quad m<\dfrac{71}{24}$

12.(D) $x^2+ax+c=(x-\gamma)(x-\delta)$

Thus, $\dfrac{(\alpha-\gamma)(\alpha-\delta)}{(\beta-\gamma)(\beta-\delta)}=\dfrac{\alpha^2+a\alpha+c}{\beta^2+a\beta+c}=\dfrac{-b+c}{-b+c}=1 \qquad [\because \alpha, \beta$ are roots of $x^2+ax+b=0$]

13.(D) Let α and 2α be the roots of (i), then $(a^2-5a+3)\alpha^2+(3a-1)\alpha+2=0$ **. . . .(i)**

and $\quad (a^2-5a+3)(4\alpha^2)+(3a-1)(2\alpha)+2=0$ **. . . .(ii)**

Multiplying (i) by (4) and subtracting it from (ii) we get : $(3a-1)(2\alpha)+6=0$

Therefore, $\alpha=-\dfrac{3}{(3a-1)}$

Putting this value in (ii) we get: $(a^2-5a+3)(9)-(3a-1)^2(3)+2(3a-1)^2=0$

$\Rightarrow \quad 9a^2-45a+27-(9a^2-6a+1)=0 \quad \Rightarrow \quad -39a+26=0 \quad \Rightarrow \quad a=\dfrac{2}{3}$

For $a=\dfrac{2}{3}$, the equation becomes $x^2+9x+18=0$, whose roots are $-3, -6$

14.(B) $\because\ f(2)$ is minimum $\qquad \therefore\ f(x)$ will be symmetric about $x=2$

$\therefore$ greater is magnitude of difference between value of x and 2, greater is the value of $f(x)$

$\therefore\ f(2)<f(0)<f(5)$

15.(D) $111x^3-11x-1=0 \qquad \therefore \qquad \alpha+\beta+\gamma=0$

$\Sigma\alpha\beta, \dfrac{-11}{111}, \alpha\beta\gamma=\dfrac{1}{111} \quad \Rightarrow \quad \dfrac{1}{(\alpha\beta)^2}+\dfrac{1}{(\beta\gamma)^2}+\dfrac{1}{(\gamma\alpha)^2}=\dfrac{\alpha^2+\beta^2+\gamma^2}{(\alpha\beta\gamma)^2}=\dfrac{(\alpha+\beta+\gamma)^2-2\Sigma\alpha\beta}{(\alpha\beta\gamma)^2}=2442$

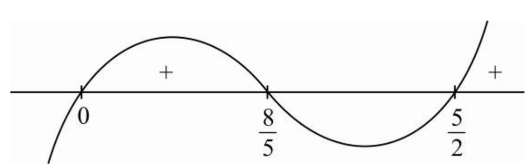
16.(B) $x^2+mx+n=0\ (\alpha,\ \beta)$**;** $\qquad \alpha+\beta=-m, \quad \alpha\beta=n$

$x^2+px+q=0\,(\alpha^3,\ \beta^3)$

$\alpha^3+\beta^3=-p \Rightarrow -p=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

$-p=(-m)^3-3n(-m)$**;** $\quad p=m^3-3mn$

17.(C) $\left|\dfrac{x^2-5x+4}{x^2-4}\right|\le 1 \qquad$ and $\quad x\ne \pm 2$



$|x^2-5x+4|\le|x^2-4|$

$(x^2-5x+4)^2\le(x^2-4)^2$

$(x^2-5x+4-x^2+4)(x^2-5x+4+x^2-4)\le 0$

$(8-5x)(2x^2-5x)\le 0$

$(5x-8)(2x^2-5x)\ge 0$

18.(D) Let $y = \dfrac{x}{x^2 - 5x + 9}$ or $x^2 y - (5y+1)x + 9y = 0$

For real x, $D \geq 0$

i.e. $(5y+1)^2 - 36y^2 \geq 0$

$\Rightarrow \quad (5y+1-6y)(5y+1+6y) \geq 0 \Rightarrow (-y+1)(11y+1) \geq 0$

$\Rightarrow \quad (y-1)(11y+1) \leq 0 \Rightarrow y \in \left[\dfrac{-1}{11}, 1\right]$

19.(C) Given inequation is $\dfrac{1}{x^2 - x + 1} \geq 4$

$\Rightarrow \quad x^2 - x + 1 \leq \dfrac{1}{4} \quad \left(\because x^2 - x + 1 = \left(x - \dfrac{1}{2}\right)^2 + \dfrac{3}{4} \geq \dfrac{3}{4} \forall x \in R\right) \quad \Rightarrow \quad x^2 - x + \dfrac{1}{4} + \dfrac{3}{4} \leq \dfrac{1}{4}$

$\Rightarrow \quad \left(x - \dfrac{1}{2}\right)^2 \leq \dfrac{1}{4} - \dfrac{3}{4} \quad \Rightarrow \quad \left(x - \dfrac{1}{2}\right)^2 \leq -\dfrac{1}{2}$ which is not true for any real x.

20.(C) Let α be a common root

$\Rightarrow \alpha^2 - \alpha - b = 0 \quad$ ...(1)

$\alpha^2 - (b+1)\alpha + 2 = 0 \quad$ ....(2)

$\alpha = \dfrac{b+2}{b}$

Putting this in eqn (1)

$\left(\dfrac{b+2}{b}\right)^2 - \left(\dfrac{b+2}{b}\right) - b = 0 \Rightarrow b^3 - 2b - 4 = 0 \Rightarrow b = 2$

SECTION - 2

21.(2) $2\alpha + 2\beta + \alpha = 20$; $\quad \alpha = -8$

$3(-8) + 2\beta = 20$; $\quad \beta = 2$

22.(8) $2x^2 + 6x + b = 0 \;\; b < 0$

$$\frac{\alpha^3 + \beta^3}{\alpha\beta} = \frac{(\alpha+\beta)\,((\alpha+\beta)^2 - 3\alpha\beta)}{\alpha\beta} = \frac{-3\left(9 - \frac{3b}{2}\right)}{\frac{b}{2}} = \frac{\left(\frac{9b}{2} - 27\right)}{\frac{b}{2}} = \frac{9(b-6)}{b} = y$$

$9b - 54 = by$; $\quad b = \dfrac{54}{9-y} > 0, \quad y - 9 < 0 \quad \Rightarrow \quad y < 9$

23.(7) $2x^2 + 5x + 7 = 0$ has non-real roots.

$\therefore$ both roots are common

$\dfrac{a}{2} = \dfrac{b}{5} = \dfrac{c}{7}$

24.(0) $y = \dfrac{x^2 - 2x + 2}{2x - 2} \Rightarrow x^2 - 2x + 2 = 2xy - 2y$

$\Rightarrow \quad x^2 - 2x(y+1) + 2(y+1) = 0$

as $x \in R$, so $D \geq 0$

$(y+1)^2 - 2(y+1) \geq 0 \Rightarrow (y+1)(y-1) \geq 0 \quad \Rightarrow y \geq 1$ or $y \leq -1$

25.(5) $x^2-(\alpha-2)x-\alpha-1=0 \begin{array}{l} \nearrow p, \\ \searrow q \end{array}$

$p^2+q^2=(p+q)^2-2pq=(\alpha-2)^2-2(-\alpha-1)=\alpha^2-4\alpha+4+2\alpha+2$

$=\alpha^2-2ga+6=(\alpha-1)^2+5$

Min value of p^2+q^2 is 5